Name:

Class: Date:

GCSE TOPIC TEST

PHYSICS

H

Higher Tier

4.2 Electricity

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
TOTAL	

Materials

For this paper you must have:

- a ruler
- a scientific calculator
- the Physics Equation sheet

Instructions

- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- In all calculations, show clearly how you work out your answer.

Information

- The maximum mark for this paper is 40.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers

01 A correctly wired plug and cable connects a washing machine to the mains electricity supply.

The washing machine has a metal case.

A fault causes the live wire to make an electrical connection with the metal case of the washing machine.

The earth wire is **not** connected to the metal case of the washing machine.

Explain why it would not be safe for a person to touch the metal case.

[2 marks]

02

A student investigated how the current in a filament lamp varied with the potential difference across the filament lamp.

Figure 1 shows part of the circuit used.

Figure 1

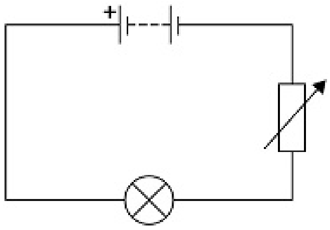
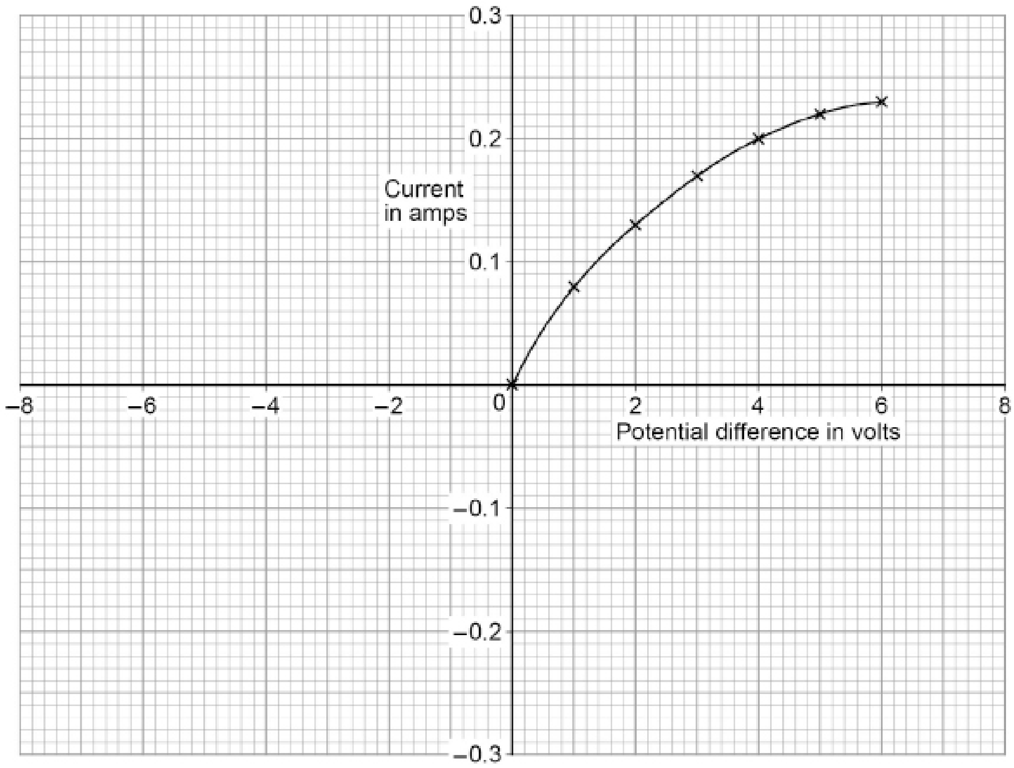


Figure 2 shows some of the results.

Figure 2



(a) Write down the equation which links current (*I*), potential difference (*V*) and resistance (*R*).

____

____

(1)

(b) Determine the resistance of the filament lamp when the potential difference across it is 1.0 V.

Use data from **Figure 2** above.

Resistance = ____________ Ohm

(4)

[5 marks]

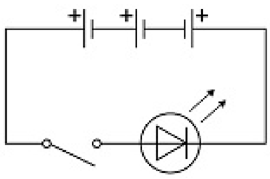
03.1 **Figure 3** shows an LED torch.

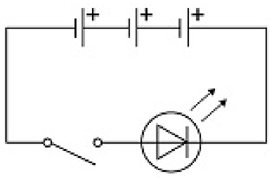
Figure 3

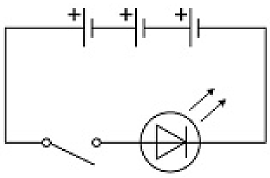
The torch contains one LED, one switch and three cells.

Which diagram shows the correct circuit for the torch?

Tick **one** box.

☐ A 

☐ B 

☐ C 

[1 mark]

03.2 **(a)** Write down the equation which links charge flow (*Q*), current (*I*) and time (*t*).

(1)

(b) The torch worked for 14 400 seconds before the cells needed replacing.

The current in the LED was 50 mA.

Calculate the total charge flow through the cells.

Total charge flow = ____________ C

(3)

[4 marks]

04 **(a)** Write down the equation which links current, potential difference and power.

(1)

A light bulb has a power input of 40 W.

The mains potential difference is 230 V.

(b) Calculate the current in the light bulb.

Current = ____________ A

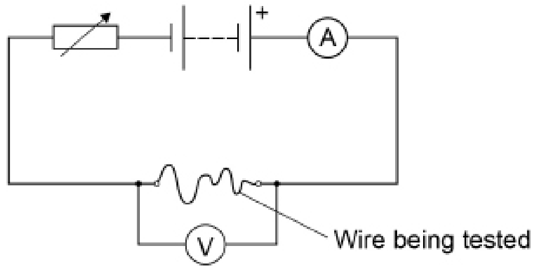
(3)

[4 marks]

05.1 A student investigated how the resistance of a piece of wire varies with its length.

Figure 4 shows the circuit used.

Figure 4



Explain why the student needed to adjust the variable resistor each time she changed the length of the wire.

____

____

____

____

____

[3 marks]

05.2 The student recorded three measurements of the potential difference across a 0.10 m length of wire.

Table 1 shows the results.

Table 1

Length in m	**Potential difference in V**			
	1	2	3	Mean
0.10	**X**	0.18	0.15	0.17

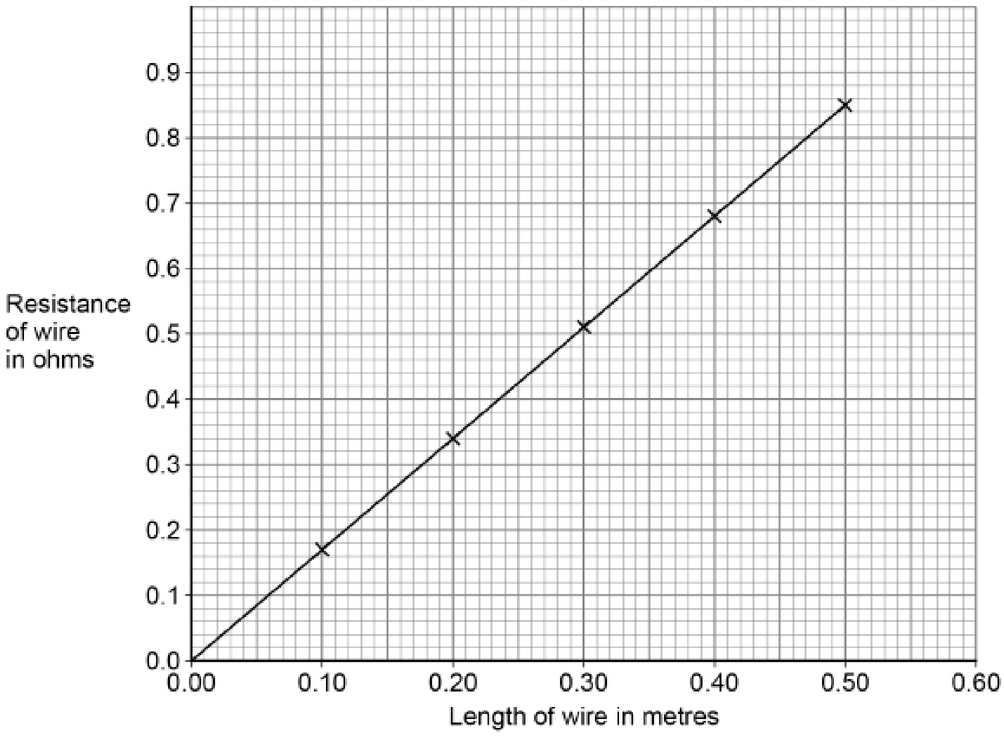
Calculate **X** in **Table 1** above.

X = ____________ V

[2 marks]

05.3 **Figure 5** shows the results for five different lengths of the wire.

Figure 5



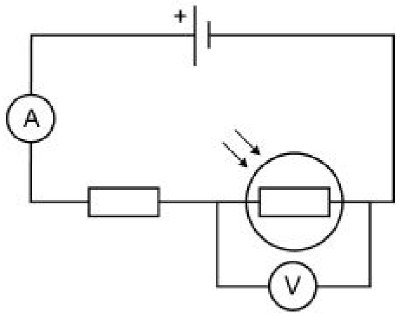
Describe the relationship between the length of the wire and the resistance of the wire.

[2 marks]

06 A student builds a circuit.

The circuit is shown in **Figure 6**.

Figure 6

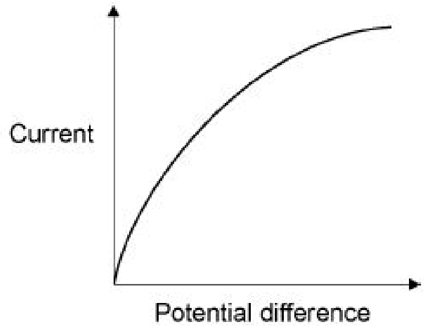


Explain how the readings on both meters change when the environmental conditions change.

[6 marks]

07 **Figure 7** shows a current – potential difference for a filament lamp.

Figure 7



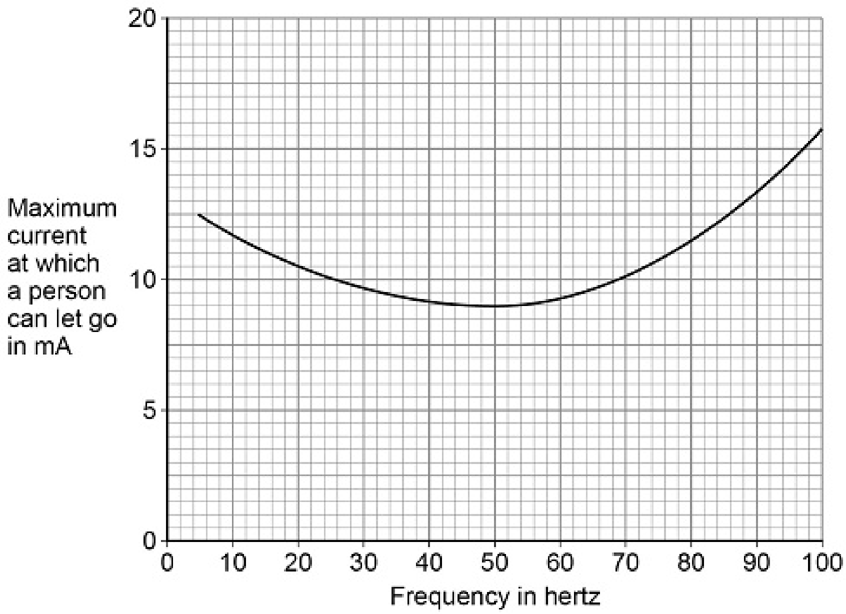
Explain how the resistance of a filament lamp changes as the potential difference across it increases.

[3 marks]

08 The current from an electric shock causes a person's muscles to contract. The person cannot let go of the electrical circuit if the current is too high.

Figure 8 shows how the maximum current at which a person can let go depends on the frequency of the electricity supply.

Figure 8



The UK mains frequency is 50 Hz.

Explain why it would be safer if the UK mains frequency was **not** 50 Hz.

[2 marks]

09 A hybrid car has an electric motor and a petrol engine.

The electric motor in the car is powered by a battery.

To charge the battery, the car is plugged into the mains supply at 230 V.

The power used to charge the battery is 6.9 kW.

Calculate the current used to charge the battery.

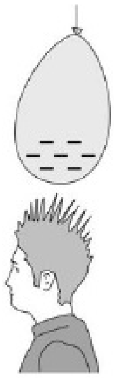
Current = ____________ A

[4 marks]

10 **Figure 9** shows a student after rubbing a balloon on his hair.

The balloon and hair have become charged.

Figure 9



Describe the force that acts on the student's hair in the diagram.

[2 marks]